STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 10: Thermal Properties of Matter

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

No net heat transfer between systems in thermal equilibrium.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q2 1 mark

300.15 K, approximately 300 K.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q3 1 mark

Heat needed to raise 1 kg by 1 K.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q4 1 mark

Q=mcDeltaT.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q5 1 mark

Q=mL.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q6 1 mark

Radiation.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q7 1 mark

pV=nRT.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q8 1 mark

Rate is approximately proportional to excess temperature for small differences.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q9 2 marks

4000 J.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q10 2 marks

1.2 x 10-^3 m.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q11 2 marks

373.15 K.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q12 2 marks

400 J/K.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q13 2 marks

66,800 J.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q14 2 marks

Energy supplies latent heat for phase change.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q15 3 marks

21,000 J.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q16 3 marks

1.2 x 10-^3 m.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q17 3 marks

44 deg C.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q18 3 marks

150 kPa.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q19 4 marks

Conduction via microscopic interactions; convection by bulk flow; radiation via electromagnetic waves.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.

Q20 4 marks

mL+mcDeltaT=83,600 J.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.